

IRTF/IETF OPS Transfer between NMRG/NMOPS

Status & Exploring Collaboration Opportunities

The NMRG charter states: “The IETF Operations and Management Area Directors are members of the NMRG mailing list and invited to NMRG meetings in order to ensure free flow of information in both directions, and to avoid duplication of work with the various IETF working groups.”

Looking at the bluesheets of IETF 125, we can observe that NMRG had 107 and NMOP had 115 participants, and **only 36 participants** were attending both NMRG and NMOP.

AI, digital twins, knowledge graphs, large language models, generative AI etc. used to be a topics for research only. Looking at the 3 ([China Unicom](#), [China Telecom](#), [China Mobile](#)) network operator presentations at the OPSAREA 125, **the gap between research and operation has considerably narrowed.**

Its perhaps time to bring these two communities closer together to.

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Is there interests and common scope?

We [reached out](#) to NMRG, NMOP and OPSAWG and asked the following questions:

- Would you agree that the lifecycle on research and industry adoption of model and data driven network management becomes accelerated and therefore research and industry adoption comes more closer together?
-> **Nobody answered no.**
- Is there an interest at the NMRG and NMOP communities to have an exchange on currently adopted and future proposed work to identify synergies and common interests in both working groups and discuss how they could be addressed better?
-> From the feedback I concluded: **YES**
- Do you have a particular topic, set of documents or interest already in mind?
-> We go **a lot of comments...**

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Example voices (1)

Brad Peters (NBN Australia): My personal views are that data driven network management is fundamental baseline as management evolves. The ability to enable AI for management processes requires not only the schema, the run time data but also the view of how these relate to the network being managed.

Bernard Kavanagh (PingCAP): yes, to all three of the questions, and Brad has articulated the reason more precisely than I've seen it put before. The stack Brad describes is right: schema, runtime data, topology view, and process encoding as an overlay for agents. What strikes me is that this is also exactly where a lot of real-world agent deployments are quietly failing, not because the information model is wrong, but because there's no defined layer for runtime retrieval and recall between that model and the agent's actual decision loop.

Christopher Janz (Huawei): But one might ask: what is the best focal point for standardization going forward? The emerging world is one in which systems are cognitive. In the pre-cognitive world, data models - a stable basis for system-to-system communication - were the natural targets for standardization. Cognitive systems may not require that basis a priori.

Iulislol Zacarias (University of Braunschweig): At EuCNC we presented the Q1/Planning and Reasoning part, where we input high-level intents, and get an "orchestrator" agent to decompose the tasks into many sub-tasks. We then measured how "well" the orchestrator agent could route the commands to the right "sub-agents" in the network.

Joe Clarke (Cisco) This is where Digital Twin comes into place. You don't necessarily want a model to "learn" as it's going through your production network. Testing its "reasoning" in a twin environment makes it safer for doing that self-correction and iteration.

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Example voices (2)

Xing Zhao (CAICT): Glad to see that many people starting to pay attention to AI agents for network management. I agree that the capabilities of AI agents include perception, planning and reasoning, tool invocation and execution, plus learning and self-optimization. I fully agree that "pre-execution validation layer" is critical and it should be done during the decision-making phase.

Dan Voyer (Cisco) I agree that the cycle between research ideas and operational adoption is getting shorter, especially around data-driven network management, anomaly detection, digital twins, and AI/agent-assisted operations. There is already relevant work that can help anchor this discussion. In NMOP, documents such as SIMAP, the network anomaly architecture and semantics work, the network incident YANG model, and YANG/message-broker integration are already touching many of these foundation

Alex Huang Feng (INSA University Lyon): I do think that both NMOP and NMRG communities have a lot to gain from a collaboration perspective. My feeling over the last few years at NMRG is that the community have been trying to standardise documents rather than being researchers and test ideas. Something I miss at NMRG is the "experiment" approach. This was very well solved at NMOP, where part of the WG meeting is solely dedicated to present Hackathon projects and other presentations related to the WG.

Qin Wu (Huawei): Yes, many AIOps related work (<https://github.com/IETF-OPS-AD/AINETOPS>) has been distributed in various different places, driven by individual, lack a good coordination, encouraging more exchange between NMRG and NMOP communities, since both Group request 2 sessions in each IETF meeting, I am wondering whether a joint meeting can be planned to make good use of time for both group members.

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a first proposal for a joint session discussion...

From what I personally observed, having an initial discussion on the following existing working group document topics and how they intersect

- Network Digital Twin ([draft-irtf-nmrg-network-digital-twin-arch](#))
- Intent-Based Networking ([draft-irtf-nmrg-ibn-usecases](#))
- SIMAP ([draft-ietf-nmop-simap-concept](#))
- YANG-Push to Message Broker integration ([draft-ietf-nmop-yang-message-broker-integration](#))
- Network Management Agent Functional Architecture ([draft-zhao-nmop-network-management-agent](#))
- Knowledge Graph Framework for Network Operations ([draft-mackey-nmop-kg-for-netops](#))

and then having a discussions on the following raised questions perhaps:

- Which AI agent's capabilities are suitable for network management?
- How does an AI agent interact with Network Digital Twin, Intent-Based Networking, SIMAP, Knowledge Graphs and YANG data in Message Brokers for which AI agent capability?
- What is the impact of an AI network management process to YANG information, service, network and data modelling?
- Is MCP (https://en.wikipedia.org/wiki/Model_Context_Protocol) a suitable protocol between AI agents and IETF network management data sources and system components? Is something missing or should be changed?

Could this be a reasonable next step. **Thoughts?**

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We just ignore that there is interest
from both sides and wait how this is progressing...

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what can we lose?

Time

-> Preparing and holding a joint session takes time.

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what can we gain?

- increasing discussions and mutual attendance count increase at NMRG and NMOP
- Documents potentially better aligned in the future
- Missing AI related gaps identified

Looking at the joint sessions between BMWG and IPPM, two working groups sharing similar charter, joint sessions proved to help in getting a common perspective.

Perhaps this works here as well? **Worth a try?**